

Bhavyashree Putta

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Professional Summary

Architected a distributed transit platform serving 65K+ users and generating \$18M+ revenue. Experienced in React, Python, and Kubernetes, combining full-stack precision with scalable, observable systems. Currently leading a Google-funded RAG chatbot using Azure OpenAI, FAISS, and Pinecone to enhance data reliability and retrieval accuracy.

Technical Skills

Languages: Python, Java, Javascript (ES6+), Typescript, C#

Frontend: React, Next.js, Redux Toolkit, HTML5, CSS3, TailwindCSS, Bootstrap

Backend: Spring Boot, Node.js, FastAPI, .NET Core, REST API, GraphQL

Data & Systems: PostgreSQL, MySQL, Redis, SQLAlchemy, ETL and Data pipelines

DevOps & Cloud: Docker, Kubernetes, Terraform, CI/CD(GitHub Actions), AWS, Microsoft Azure

Testing & QA: React Testing Library (RTL), Jest, Cypress, Playwright

Coursework: Data structures & Algorithms, Distributed Cloud Computing, Database systems, Artificial Intelligence

Experience

UIC College of Applied Health Sciences | Research Software Engineer

June 2025 - Present

- Built and deployed a **Google-funded RAG chatbot** with GPT-4, Pinecone, and FAISS, enabling accurate retrieval with a **1.2s median latency** even in low-connectivity settings.
- Integrated Twilio WhatsApp for real-time chats and sensitive-query escalation, engineered Cloudflare D1 + KV session storage to support **50+ concurrent users with 99.9% uptime** and zero session loss.
- Containerized microservices using Docker, automated IaC with Terraform, and managed deployments through Kubernetes + Helm, **achieving 60% fewer response failures** and improved observability across pods.

University of Illinois Chicago | Graduate Assistant - Fullstack Developer

Aug 2023 - May 2025

- Designed and scaled a full-stack transit platform for 65K+ users, optimized SPA for low-end devices, cutting load times by 45% and sustaining **p99 latency under 120ms with 99.95% uptime**.
- Developed production-grade REST APIs in Python and C# with schema validation and async workflows, added Pytest/xUnit suites that **cut integration bugs 40%** and stabilized deployments.
- Automated an **Python ETL pipeline** handling 50K+ daily enrollment data with eligibility checks and Azure uploads, **reducing manual effort by 90%** and **increasing data accuracy by 30%**.
- Optimized SQL databases for transit and payment systems with ACID transactions, migrations, and indexing, **boosting query performance by 60%** and enabling zero-downtime releases.
- Rebuilt UIC departmental sites with WCAG 2.1-AA accessibility, optimizing HTML/CSS and static caching to **reduce average load time by 35%** and enhance usability based on student feedback.

GetLect | Web Developer

May 2022 - Apr 2023

- Implemented 25+ modular React components from Figma designs using TypeScript, HTML5, and CSS3, accelerating design-to-dev handoffs by 40% and **reducing UI revision cycles by 30%** across sprints.
- Refactored a legacy monolithic backend into modular Spring Boot services and migrated workloads to AWS (EC2, S3), **cutting on-prem servers by 30%** and saving \$8K/year in recovery costs.
- Deployed RESTful APIs behind Nginx load balancers with active health checks and rate limiting, maintaining consistent latency and **achieving 99.9% uptime during peak traffic**.

Projects

Receiptaurant | React, TypeScript, Node.js, Express, MySQL, AWS, Redis

- Engineered a real-time AI-powered surcharge detection system with REST and GraphQL APIs, tuned Google Studio LLM prompts, and **cached results in Redis for sub-200 ms responses**.
- Developed interactive D3.js + Google Maps heatmaps with **MySQL (AWS RDS) and S3-backed receipt storage**, visualizing 100's of parsed receipts and strengthening skills in data visualization and cloud architecture.

Valorant Esports Manager | Amazon Bedrock, Pinecone, FastAPI, React.js, Riot API, Python

- Built an LLM-powered scouting assistant on Amazon Bedrock (Claude 3 Haiku) for **real-time player analysis**, lineup suggestions, and recruiting insights across **119,000 professional matches**.
- Engineered a FastAPI backend with Pinecone vector search to retrieve **1,536-dimensional player embeddings**, parallelizing Riot API ingestion to support high-throughput queries over the full dataset.

Education

University of Illinois Chicago, USA

Aug 2023 – May 2025

Master of Science in Computer Science

GPA 3.88

- Featured by **Cook County Chronicle** for leading U-PASS+, expanding transit access to over 65,000 students.
- Won **Epic Cosmos Hackathon** for a data-driven HRT-breast cancer analysis using Cosmos data.